

Candidate Elimination Algorithm

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import numpy as np

concepts = np.array(data.iloc[:, :-1])

target = np.array(data.iloc[:, -1])

S = concepts[0].copy()

G = [['?' for i in range(len(S))] for i in range(len(S))]

for i in range(len(concepts)):
    if target[i] == "Yes":
        for j in range(len(S)):
            if concepts[i][j] != S[j]:
                S[j] = '?'
                G[j][j] = '?'

    else:
        for j in range(len(S)):
            if concepts[i][j] != S[j]:
                G[j][j] = S[j]
            else:
                G[j][j] = '?'

print("\nSpecific Hypothesis:")
print(S)

print("\nGeneral Hypothesis:")

for i in G:
    if i != ['?'] * len(S):
        print(i)
```

```
...
Initial Specific Hypothesis:
['D1' 'Overcast' 'Mild' 'Normal' 'Strong']

Initial General Hypothesis:
[['?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?']]

Final Specific Hypothesis:
['?' '?' '?' '?' '?']
```